SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1023	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:		**Reg. No.:**	
Date:		**Duration:**	60 minutes

Code of Conduct

I M. Varun Sai (192425247_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M. Varun Sai_

Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

1. **Understand the Problem Statement**
   - Explain the problem in your own words.
   - Identify the objectives of the proposed system.
   - State the scope of the problem.
2. **Identify Key Stakeholders and Challenges**
   - Identify all the stakeholders involved in the system.
   - Explain the roles and responsibilities of each stakeholder.
   - Discuss the major challenges and issues faced by the stakeholders.
3. **Analyze the Current Approach**
   - Describe the existing system or current process.
   - Identify its strengths and limitations.
   - Analyze the problems or gaps that need to be addressed.
4. **Propose a Suitable Solution Using Software Engineering Principles**
   - Design a software-based solution for the given problem.
   - Describe the major functional modules of the proposed system.
   - Identify the functional and non-functional requirements.
   - Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.
5. **Justify the Chosen Methodology/Framework**

- Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).
- Justify why the selected methodology/framework is suitable for the proposed system.
- Explain how it supports successful project development and maintenance.

6. **Discuss Expected Outcomes and Limitations**
   - Describe the expected benefits of the proposed solution.
   - Explain how the solution addresses the identified challenges.
   - Discuss potential risks, implementation challenges, and limitations.
   - Suggest possible future enhancements.

Intelligent E-Waste Collection and Sustainable Recycling Platform

1. Understand the Problem Statement

1.1 Problem Explanation

The rapid use of computers, mobile phones, batteries, televisions, and other electronic devices has resulted in a large increase in **e-waste**. Existing collection and recycling processes are often disconnected and depend on manual coordination.

The proposed **Intelligent E-Waste Collection and Sustainable Recycling Platform** provides a centralized system that connects consumers, collection centres, recyclers, logistics providers, and administrators. It allows users to request e-waste collection, schedule pickups, track materials, record recycling activities, and monitor sustainability indicators.

1.2 Objectives

The main objectives are:

- To provide an online platform for e-waste collection requests.
- To schedule and track e-waste pickups.
- To classify e-waste based on material or device type.
- To maintain recycling and recovery records.
- To provide sustainability dashboards and reports.
- To reduce manual coordination between stakeholders.
- To provide secure and scalable APIs.
- To use containerization, monitoring, testing, and CI/CD.
- To encourage responsible disposal and recycling of electronic devices.

1.3 Scope

The system covers:

- Consumer registration and login.
- E-waste collection requests.
- Pickup scheduling and logistics management.
- E-waste classification.
- Collection and transportation tracking.
- Recycling and material recovery records.
- Recycler and collection-centre management.
- Sustainability analytics and reports.
- Admin management.
- Secure APIs and database management.
- Monitoring, logging, testing, and automated deployment.

2. Identify Key Stakeholders and Challenges

2.1 Stakeholders

Stakeholder	Role and Responsibility
Consumers	Register e-waste, request pickups, select pickup locations and track requests.
Collection Centres	Receive, store, weigh, classify, and transfer e-waste.
Recyclers	Process e-waste and recover reusable materials.
Logistics Providers	Collect and transport e-waste between locations.
Administrators	Manage users, centres, recyclers, requests, and system operations.
Government/Regulatory Bodies	Ensure compliance with e-waste and environmental regulations.
System Developers	Develop, test, maintain, and improve the platform.
System Operators/Monitoring Team	Monitor system performance, logs, availability, and security.

2.2 Major Challenges

Consumers

- Lack of awareness about proper e-waste disposal.
- Difficulty finding nearby collection facilities.
- Lack of information about pickup status.

Collection Centres

- Manual record maintenance.
- Difficulty managing large quantities of e-waste.
- Poor visibility of incoming and outgoing materials.

Recyclers

- Difficulty tracking material sources.
- Inconsistent classification of e-waste.
- Need for accurate recycling and recovery records.

Logistics Providers

- Pickup coordination problems.
- Inefficient transportation routes.
- Difficulty tracking material movement.

Administrators

- Managing multiple stakeholders.
- Maintaining accurate data.
- Monitoring the complete e-waste lifecycle.
- Ensuring system security and availability.

3. Analyze the Current Approach

3.1 Existing System

The current e-waste process is commonly fragmented:

Consumer → Local Collector → Collection Centre → Transporter → Recycler → Material Recovery

Many activities may be performed through phone calls, paper records, spreadsheets, or separate systems.

3.2 Strengths

- Simple to operate on a small scale.
- Existing collection centres can be used.
- Requires limited technical infrastructure.
- Local collectors can provide direct services.

3.3 Limitations

- Lack of centralized information.
- Manual collection and record management.
- Poor tracking of e-waste movement.
- Difficulty coordinating pickups.
- Limited recycling analytics.
- Higher possibility of data errors.
- Limited visibility for consumers.
- Difficult to scale when the number of users increases.

3.4 Problems/Gaps

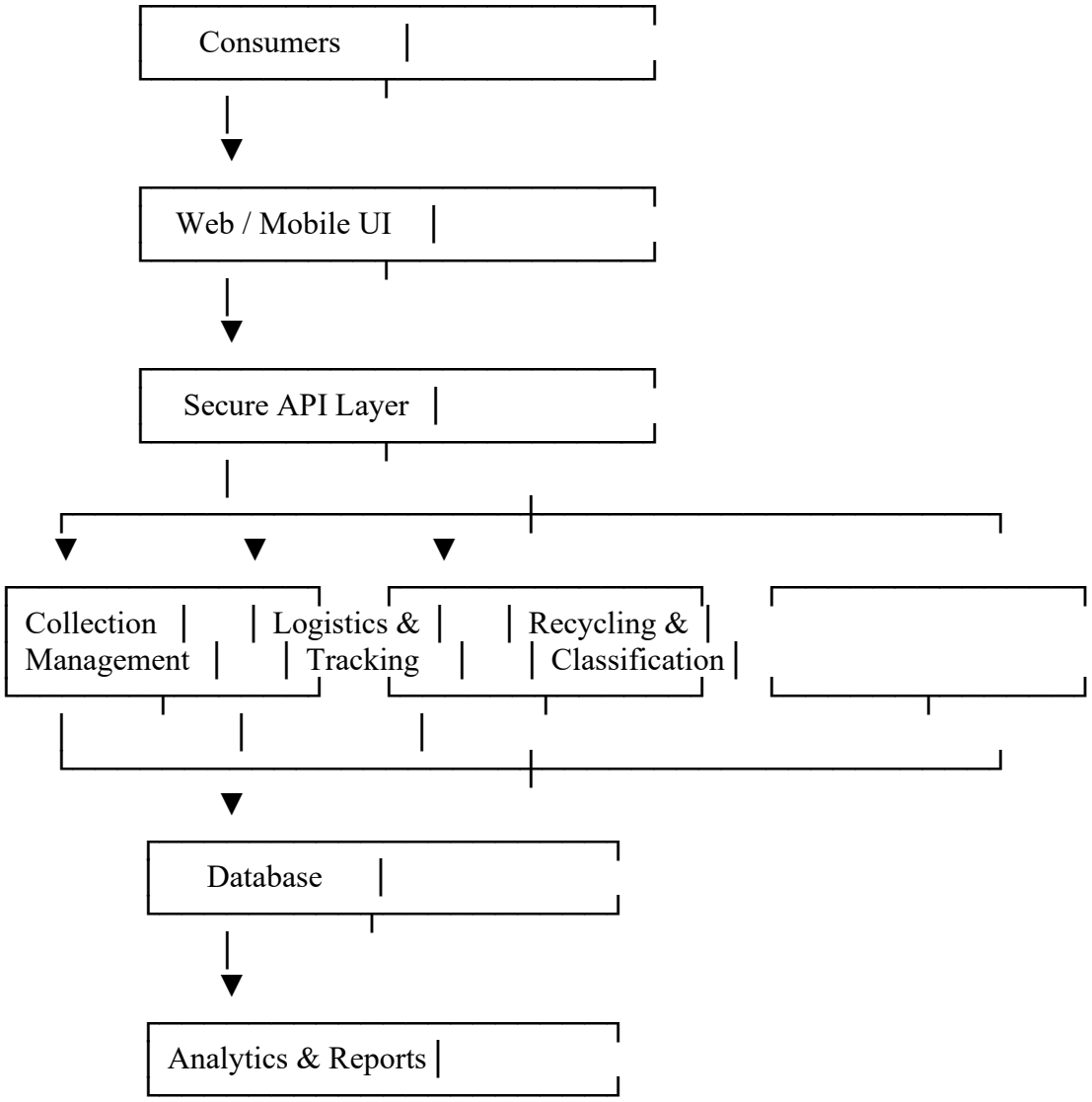
The major gap is the absence of an **integrated digital platform** connecting all stakeholders.
The proposed system addresses this by providing:
Collection → Classification → Pickup → Transportation → Recycling → Recovery → Analytics
within one platform.

4. Proposed Solution Using Software Engineering Principles

4.1 Proposed System

The proposed platform is a web-based/cloud-based system that provides a centralized interface for consumers, collection centres, logistics providers, recyclers, and administrators.

Basic Architecture



Monitoring + Logging + CI/CD + Containers

4.2 Major Functional Modules

Module 1: User and Request Management

Purpose: Manage consumers and their e-waste collection requests.

Functions:

- User registration and login.
- Add e-waste details.
- Create collection requests.
- View request status.
- Cancel or modify requests.

Output: Collection request and status information.

Module 2: E-Waste Classification

Purpose: Categorize collected electronic waste.

Functions:

- Record device type.
- Record quantity and weight.
- Classify recyclable, reusable, or hazardous materials.
- Store classification records.

Output: Classified e-waste information.

Module 3: Pickup and Logistics Management

Purpose: Coordinate collection and transportation.

Functions:

- Assign pickup requests.
- Schedule collection.
- Assign logistics providers.
- Track transportation status.
- Update delivery information.

Output: Pickup and transportation status.

Module 4: Recycling Management

Purpose: Track recycling and material recovery.

Functions:

- Record recycler information.
- Record recycling processes.
- Track recovered materials.
- Maintain recycling history.

Output: Recycling and recovery records.

Module 5: Sustainability Analytics

Purpose: Measure the environmental impact of recycling activities.

Functions:

- Calculate total e-waste collected.
- Track recycled and recovered materials.
- Generate sustainability reports.
- Display dashboards and statistics.

Output: Sustainability indicators and reports.

Module 6: Administration and Monitoring

Purpose: Manage and monitor the complete platform.

Functions:

- Manage users and stakeholders.
- Monitor system activity.

- Manage permissions.
- View logs and errors.
- Monitor API and service performance.

Output: Administrative reports and system status.

4.3 Functional Requirements

The system shall:

1. Allow users to register and authenticate.
2. Allow consumers to submit e-waste collection requests.
3. Allow pickup scheduling.
4. Allow logistics providers to update delivery status.
5. Allow collection centres to classify e-waste.
6. Allow recyclers to record recycling activities.
7. Track e-waste throughout its lifecycle.
8. Generate sustainability reports.
9. Provide administrator dashboards.
10. Provide secure APIs for communication between services.

4.4 Non-Functional Requirements

Requirement	Description
Security	Authentication, authorization, encryption, and secure APIs.
Scalability	Support increasing users, requests, and transactions.
Reliability	Provide consistent and dependable services.
Performance	Respond quickly to user and API requests.
Usability	Provide a simple and easy-to-use interface.
Maintainability	Allow individual modules to be updated easily.
Availability	Keep important services available with minimum downtime.
Monitoring	Track system health, errors, and service performance.
Portability	Support deployment using containers.

5. Software Engineering Principles

Modularity

The system is divided into independent modules such as collection, logistics, recycling, and analytics.

Scalability

Containerized services and scalable APIs allow the system to support increasing users and transactions.

Maintainability

A modular architecture makes it easier to modify or replace individual components without affecting the entire system.

Reliability

Database backups, error handling, monitoring, logging, and testing improve system reliability.

Usability

Simple dashboards and clear status information make the platform easy for consumers and other stakeholders.

Security

Role-based access control, secure authentication, API security, encryption, and input validation protect user and system data.

Sustainable Software Practices

The system can reduce unnecessary resource consumption through efficient APIs, optimized database queries, lightweight services, and cloud-resource monitoring.

6. Justify the Chosen Methodology/Framework

Selected Methodology: Agile + DevOps

Agile is suitable for developing the platform because it contains multiple modules and involves different stakeholders with changing requirements.

Why Agile?

- Requirements can be developed incrementally.
- Stakeholders can provide regular feedback.
- Modules can be developed in short iterations.
- Problems can be identified early.
- New features can be added based on user requirements.

DevOps Integration

DevOps can be used for:

- Automated testing.
- Continuous Integration.
- Continuous Deployment.
- Containerization using Docker.
- Application monitoring.
- Centralized logging.
- Automated builds and deployments.

Development Process

Requirement Analysis
↓
Sprint Planning
↓
Design & Development
↓
Testing
↓
Continuous Integration
↓
Containerized Deployment
↓
Monitoring
↓
User Feedback
↓
Next Sprint

This combination supports continuous improvement and reliable deployment of the platform.

7. Expected Outcomes and Limitations

7.1 Expected Benefits

The proposed system is expected to:

- Improve visibility of e-waste movement.
- Increase recycling and material recovery.
- Reduce manual coordination.
- Improve pickup efficiency.
- Reduce loss or misplacement of materials.
- Provide accurate recycling records.
- Support sustainability reporting.
- Improve communication between stakeholders.

- Encourage responsible e-waste disposal.
- Provide better decision-making through analytics.

7.2 How the Solution Addresses Challenges

Existing Challenge	Proposed Solution
Manual requests	Online collection requests
Poor tracking	Digital status and movement tracking
Pickup coordination	Automated scheduling and assignment
Classification difficulties	Digital classification records
Poor recycling visibility	Recycling lifecycle tracking
Lack of sustainability data	Analytics dashboard
Data security problems	Authentication and authorization
Difficult scaling	APIs and containerized services
System failures	Monitoring, logging, and automated testing

7.3 Risks and Limitations

- Initial development and deployment cost.
- Dependence on internet connectivity.
- Users may have limited awareness of the platform.
- Incorrect information may be entered by users.
- Integration with existing recycling centres may be difficult.
- Security threats such as unauthorized access may occur.
- Logistics tracking may be inaccurate if providers do not update information.
- Large-scale deployment requires cloud infrastructure and maintenance.

7.4 Future Enhancements

Future versions can include:

- **AI-based e-waste image classification** using photographs.
- **GPS-based real-time vehicle tracking.**
- **AI-based pickup route optimization.**
- **IoT sensors** for monitoring storage conditions and container capacity.
- **Blockchain-based recycling traceability.**
- **Mobile application** for consumers and collection agents.
- **Reward system** for users who responsibly recycle e-waste.
- **Predictive analytics** to forecast e-waste generation.
- Integration with government environmental reporting systems.
- Carbon-footprint calculation and automated sustainability certificates.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding (30%)	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis (25%)	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis (20%)	Thoroughly analyzes the existing system, strengths,	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are
	weaknesses, and identifies all gaps.			largely missing.
Proposed Software Solution & Software Engineering Principles (15%)	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.
Methodology/ Framework Justification (10%)	Selects the most suitable SDLC/model/frame work and provides strong technical justification.	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing; report lacks organization and clarity.

Criteria	Mark
Problem Understanding (30%)	/5
Stakeholder & Challenge Analysis (25%)	/5
Current System Analysis (20%)	/5
Proposed Software Solution & Software Engineering Principles (15%)	/5
Methodology/ Framework Justification (10%)	/5
TOTAL	/25